

Soccer TIPS Engineering Department

Document: Layer 2_v0.2 General Developer Handoff Notes

General Developer

Layer 2 v0.1 Feature Engineering Foundation + Layer 2 v0.2 Football Sanity Validation Handoff Notes

Finalized after local Layer 2 v0.2 football sanity QA run. This document records the accepted Layer 2 v0.1 foundation and the completed Layer 2 v0.2 football sanity validation checkpoint.

1. Current Layer 2 v0.1 Status

- Layer 2 v0.1 is accepted as a successful feature-engineering foundation checkpoint.
- The active provider for Layer 2 v0.1 is SkillCorner because it has validated phases_of_play data with in-possession and out-of-possession tactical labels.
- The implementation consumes Layer 1 canonical SkillCorner outputs and writes reusable Layer 2 feature tables under data/processed/features/.
- It produces frame-level team shape metrics, phase-segment summaries, match-level tactical profiles, and a Layer 2 QA summary JSON.
- It supports both one-match processing and controlled batch processing across all processed SkillCorner matches.
- Schemas, metric sanity checks, batch processing, and phase-linking QA pass for all 10 processed SkillCorner matches.
- Orientation normalization exists only as a conservative/inactive module for now. Current analysis coordinates use the existing Layer 1 metric coordinates.
- This handoff does not authorize Layer 3 modeling, AI interpretation, dashboards, reports, recommendations, or advanced tactical claims.

2. Files Created

- src/soccer_tips/features/__init__.py
- src/soccer_tips/features/layer2_schemas.py
- src/soccer_tips/features/phase_context.py
- src/soccer_tips/features/orientation.py
- src/soccer_tips/features/team_shape.py
- src/soccer_tips/features/aggregation.py
- src/soccer_tips/features/build_layer2.py
- src/soccer_tips/features/qa_layer2_outputs.py
- tests/test_layer2_features.py

3. What Each File Does

File	Purpose
features/__init__.py	Defines the Layer 2 features package.
layer2_schemas.py	Defines schemas and schema helpers for frame team shape metrics, phase segment summaries, match tactical

	profiles, and Layer 2 QA output.
phase_context.py	Links SkillCorner Layer 1 tracking frames to phases_of_play.csv using frame_start, frame_end, and period. It determines in-possession versus out-of-possession status for each team-frame row.
orientation.py	Provides conservative analysis-coordinate helpers. Attacking-direction normalization is intentionally inactive in v0.1.
team_shape.py	Computes frame-level team shape metrics from x_metric/y_metric coordinates.
aggregation.py	Aggregates frame-level metrics into phase-segment summaries and match-level tactical profiles.
build_layer2.py	Command-line runner for one-match or batch Layer 2 feature generation.
qa_layer2_outputs.py	Checks output existence, schemas, metric sanity, and phase-linking behavior. Writes layer2_v01_qa_summary.json.
tests/test_layer2_features.py	Lightweight sanity tests for schema definitions, shape metric calculations, convex hull safety, and phase-linking examples.

4. How to Run Layer 2 v0.1

Open the terminal from the project root folder. On the current Windows setup, that project root is:

```
C:\Users\amsch\. Personal Work\Projects\Soccer TIPS
```

Commands below assume the VS Code integrated PowerShell terminal from the project root.

Task	PowerShell command
Activate virtual environment, if applicable	.\venv\Scripts\Activate
Install requirements	pip install -r requirements.txt
Set PYTHONPATH for src-layout imports	\$env:PYTHONPATH="src"
Build one SkillCorner match	python -m soccer_tips.features.build_layer2 --provider skillcorner --skillcorner-match-id 2017461
Build all processed SkillCorner matches	python -m soccer_tips.features.build_layer2 --provider skillcorner --batch

Recommended Run Order

1. Open PowerShell from the Soccer TIPS project root.
2. Activate the virtual environment if the project uses one.
3. Install requirements if needed.
4. Set PYTHONPATH to src.
5. Run the Layer 2 tests.
6. Run one SkillCorner match first.
7. Run the Layer 2 QA utility.
8. Run full SkillCorner batch mode.
9. Run the Layer 2 QA utility again.

```
.\venv\Scripts\Activate
```

```

pip install -r requirements.txt
$env:PYTHONPATH="src"
python tests/test_layer2_features.py
python -m soccer_tips.features.build_layer2 --provider skillcorner --skillcorner-match-id 2017461
python -m soccer_tips.features.qa_layer2_outputs
python -m soccer_tips.features.build_layer2 --provider skillcorner --batch
python -m soccer_tips.features.qa_layer2_outputs

```

5. How to Run Layer 2 Tests

```

$env:PYTHONPATH="src"
python tests/test_layer2_features.py

```

Expected result:

All Layer 2 feature sanity checks passed.

6. How to Run Layer 2 QA

```

$env:PYTHONPATH="src"
python -m soccer_tips.features.qa_layer2_outputs

```

Expected batch result after the accepted checkpoint:

Status: pass

7. Output Locations

Output	Location
Layer 2 feature outputs	data/processed/features/
Frame-level team shape metrics	data/processed/features/frame_team_shape_metrics.csv
Phase-segment summaries	data/processed/features/phase_segment_summaries.csv
Match tactical profiles	data/processed/features/match_tactical_profiles.csv
Layer 2 QA summary JSON	data/processed/features/layer2_v01_qa_summary.json

8. Output File Descriptions and Exact Row Grain

Output file	Description	Exact row grain
frame_team_shape_metrics.csv	Frame-level team shape metrics with phase context where available.	One row per provider, match_id, frame, and team_id. For SkillCorner v0.1 this generally means two team rows per usable tracking frame. The row represents a team-phase-frame snapshot. phase_index and phase label columns are populated only when the frame falls inside a SkillCorner phase interval.
phase_segment_summaries.csv	Aggregated summary of frame-level	One row per provider, match_id,

	team metrics for phase segments.	phase_index, period, team_id, team_in_possession_id, in-possession phase label, out-of-possession phase label, and possession_status_for_team combination that has at least one linked frame-team metric row.
match_tactical_profiles.csv	Long-form match-level tactical profile table derived from frame-level metrics.	One row per provider, match_id, team_id, and profile scope. Current profile scopes include overall, possession_status, in_possession_phase, and out_of_possession_phase. Phase-specific profile rows are grouped by the relevant possession status and phase type.
layer2_v01_qa_summary.json	Structured QA report for the generated Layer 2 outputs.	One JSON object per QA run. It contains file checks, schema checks, metric checks, phase-linking diagnostics, warnings, and errors.

9. Metrics Currently Computed

- centroid_x: mean x_metric location of the team players used in the frame.
- centroid_y: mean y_metric location of the team players used in the frame.
- team_width: max y_metric minus min y_metric for the team in the frame.
- team_depth: max x_metric minus min x_metric for the team in the frame.
- convex_hull_area: area of the convex hull around team player coordinates when at least 3 valid player points exist.
- avg_distance_to_centroid: average Euclidean distance from each used player to the team centroid.
- player_count_used: number of valid player coordinate rows used for the team-frame calculation.
- detected_player_count: count of players with is_detected == True.
- extrapolated_player_count: count of players with is_detected == False.
- metric_warning_flag: Boolean flag for metric-quality warnings generated by the shape calculation layer.

10. Batch QA Results

Check	Accepted checkpoint result
Layer 2 tests	Pass
Single-match SkillCorner build for 2017461	Pass
Single-match Layer 2 QA	Pass
Batch SkillCorner build	Pass across all 10 processed SkillCorner matches
Batch Layer 2 QA	Pass
frame_team_shape_metrics.csv	873,296 rows x 21 columns, schema pass
phase_segment_summaries.csv	9,104 rows x 46 columns, schema pass
match_tactical_profiles.csv	394 rows x 21 columns, schema pass
negative_team_width_count	0
negative_team_depth_count	0
invalid_convex_hull_count	0
metric_warning_count	0
phase_linking status	Pass

linked_phase_row_count	609,614
missing_phase_context_count	263,682 rows
phase_linking_coverage_rate	Approximately 0.698
missing_inside_phase_interval_frame_count	0
missing_outside_phase_interval_frame_count	131,841 frames

11. Phase-Linking QA Interpretation

- The lower raw phase-linking coverage rate is not a linker failure.
- The key QA finding is that frames inside actual SkillCorner phase intervals had 0 failed links.
- Across the batch, missing_inside_phase_interval_frame_count was 0.
- The missing phase context rows are explained by tracking frames outside actual SkillCorner phase intervals, mostly gaps between phase windows.
- The QA utility was updated so it does not treat low raw phase coverage as a warning when all frames inside actual phase intervals link correctly.
- The current QA interpretation is: phase linking passes when no missing frame falls inside an actual SkillCorner phase interval. Low raw phase coverage can be acceptable when caused by gaps between phase intervals.
- Frame-level rows with missing phase context should not be used for phase-segment interpretation unless future logic explicitly assigns or fills phase context.

12. Known Limitations

- Layer 2 v0.1 is a feature-engineering foundation, not a completed tactical model.
- SkillCorner is the only active Layer 2 v0.1 provider for phase-aware processing.
- Metrica is intentionally not included in phase-aware Layer 2 v0.1 batch processing because it lacks the same validated phases_of_play structure.
- Orientation normalization is not active. The orientation module is conservative and currently preserves Layer 1 metric coordinates as analysis coordinates.
- Phase labels are only attached where SkillCorner phases_of_play intervals cover the tracking frame.
- The current outputs overwrite the existing Layer 2 output CSVs when the build runner is executed.
- The match_tactical_profiles.csv output is match-level and long-form; it is not yet a final cross-match team fingerprint dataset.
- No visual validation or football interpretation of the metric values has been completed yet.

13. What Remains Intentionally Out of Scope

- Layer 3 modeling.
- Clustering or unsupervised tactical identity modeling.
- AI interpretation or prompt-based tactical analysis.
- Dashboards.
- Reports or automated report generation.
- Recommendation systems.
- Advanced compactness interpretation.
- Transition stability modeling.
- Final tactical identity claims.

- Attacking-direction normalization.
- Cross-match team-level tactical fingerprints as a final analytical product.

14. Future Code-Helper Usage Note

Future code helpers should use this document together with:

- Engineering Department Doc, Validation & Data Ingestion Handoff Summary.
- Engineering Department Doc, Layer 1_v0.1 General Developer Handoff Notes.
- AIB Project System Architecture / Workflow Specification.
- This document explains what was built for Layer 2 v0.1 and how to run it.
- The validation handoff explains why the ingestion and phase-linking rules exist.
- The Layer 1 handoff explains how canonical processed inputs are produced.
- The architecture document explains how Layer 2 connects to the full Soccer TIPS system.
- Future helpers should not use this handoff alone to authorize Layer 3 modeling, AI interpretation, dashboards, reports, recommendations, or advanced tactical claims.

15. Scope Guard

- This handoff documents the accepted Layer 2 v0.1 feature-engineering foundation only.
- Do not add new Layer 2 metrics from this document alone.
- Do not begin compactness interpretation or transition stability modeling from this document alone.
- Before expanding metrics, use this document with the architecture document and obtain Lead Developer direction.
- Before moving beyond feature engineering into tactical modeling or interpretation, obtain Lead Developer and Project Manager review.

File	Purpose
football_sanity.py	Builds football sanity summaries from existing Layer 2 outputs. It compares current shape metrics by SkillCorner phase label and runs reasonableness checks for current metrics.
visualize_layer2.py	Creates validation-only pitch plots and phase-segment time-series plots. These plots are analytical sanity checks, not dashboards or final reports.
qa_football_sanity.py	Command-line runner for Layer 2 v0.2 football sanity validation. It writes CSV and JSON summaries and creates selected validation figures.
tests/test_football_sanity.py	Lightweight tests for the football sanity helper logic and validation checks.
requirements.txt	Includes matplotlib so the project can generate validation plots.
Task	PowerShell command
Activate virtual environment, if applicable	.\venv\Scripts\Activate
Install or update requirements	pip install -r requirements.txt
Set PYTHONPATH for src-layout imports	\$env:PYTHONPATH="src"

File	Purpose
Run Layer 2 football sanity tests	python tests/test_football_sanity.py
Rebuild Layer 2 batch outputs, if needed	python -m soccer_tips.features.build_layer2 --provider skillcorner --batch
Run Layer 2 v0.2 football sanity validation	python -m soccer_tips.features.qa_football_sanity
Output	Location
Football sanity CSV summary	data/processed/features/layer2_v02_football_sanity_summary.csv
Football sanity JSON summary	data/processed/features/layer2_v02_football_sanity_summary.json
Validation figures	outputs/figures/layer2_sanity/
Check	Acceptable result
Tests	tests/test_football_sanity.py passes.
Required outputs	CSV summary, JSON summary, and selected validation plots are created.
Reasonableness checks	No fail-level metric reasonableness issues unless documented and accepted by the Lead Developer.
Visual checks	Pitch plots show plausible team separation, player locations inside expected pitch bounds, sensible centroids, valid hulls where enough players exist, and no obvious coordinate or team-id corruption.
Phase time-series checks	Selected phase segments show plausible movement over time without obvious impossible jumps caused by data or linking errors.
Interpretation status	The JSON summary should clearly distinguish pass, warning, and fail findings. Warnings can be acceptable if they are explainable and documented.

16. Layer 2 v0.2 Football Sanity Validation Addendum

Purpose: This addendum documents the completed Layer 2 v0.2 football sanity validation workflow. The goal of this checkpoint was not to add new tactical metrics, but to validate whether the existing Layer 2 v0.1 metrics behave plausibly as football data.

Final local QA status: warning. This is not an engineering failure. The workflow ran successfully and produced the expected CSV, JSON, and validation figures. The remaining warning is a football-sanity caution on `team_width_pitch_bound` that should be reviewed before using `team_width` for stronger tactical claims.

Scope guard: This addendum does not authorize Layer 3 modeling, clustering, tactical identity modeling, AI interpretation, dashboards, reports, recommendations, advanced compactness interpretation, transition stability modeling, or new metric expansion.

16.1 Files Created or Changed for Layer 2 v0.2

- `src/soccer_tips/features/football_sanity.py`
- `src/soccer_tips/features/visualize_layer2.py`
- `src/soccer_tips/features/qa_football_sanity.py`

- src/soccer_tips/features/__init__.py
- tests/test_football_sanity.py
- requirements.txt, updated to include matplotlib for validation plots if it was not already present.

16.2 What Each New File Does

File	Purpose
football_sanity.py	Builds the football sanity validation workflow. It summarizes existing Layer 2 metrics by major SkillCorner phase labels, runs reasonableness checks, and writes structured CSV/JSON validation summaries.
visualize_layer2.py	Creates validation-only pitch plots and phase-segment time-series plots. These are analytical QA visuals, not dashboard/reporting deliverables.
qa_football_sanity.py	Command-line entry point for Layer 2 v0.2 football sanity QA. It writes layer2_v02_football_sanity_summary.csv and layer2_v02_football_sanity_summary.json and saves selected figures.
features/__init__.py	Exposes the new football sanity modules as part of the features package.
tests/test_football_sanity.py	Lightweight tests for the football sanity helpers, reasonableness status classification, and expected output structure.
requirements.txt	Adds matplotlib support for validation plot generation if it was not already included.

16.3 How to Run Layer 2 v0.2 Football Sanity Validation

Open the terminal from the Soccer TIPS project root. On the current Windows setup, that is:

C:\Users\amsch\ Personal Work\Projects\Soccer TIPS

Commands below assume the VS Code integrated PowerShell terminal from the project root.

Task	PowerShell command
Activate virtual environment, if applicable	.\venv\Scripts\Activate
Install or update requirements	pip install -r requirements.txt
Set PYTHONPATH for src-layout imports	\$env:PYTHONPATH="src"
Run football sanity tests	python tests/test_football_sanity.py
Confirm/update Layer 2 batch outputs	python -m soccer_tips.features.build_layer2 --provider skillcorner --batch
Run football sanity QA	python -m soccer_tips.features.qa_football_sanity

Recommended run order:

1. Activate the virtual environment.
2. Install or update requirements.
3. Set PYTHONPATH to src.

4. Run tests/test_football_sanity.py.
5. Run or confirm the SkillCorner Layer 2 batch output.
6. Run python -m soccer_tips.features.qa_football_sanity.
7. Review the terminal output, JSON summary, CSV summary, and generated plots before treating the checkpoint as complete.

16.4 New Output Locations

Output	Location
Football sanity CSV summary	data/processed/features/ layer2_v02_football_sanity_summary.csv
Football sanity JSON summary	data/processed/features/ layer2_v02_football_sanity_summary.json
Validation figures	outputs/figures/layer2_sanity/

16.5 Final Football Sanity QA Result

QA item	Final local result
Overall status	warning
Matches inspected	10
Frame-team rows inspected	873,296
Phase segment rows inspected	9,104
Match profile rows inspected	394
Phases inspected	build_up, create, defending_transition, finish, high_block, low_block, medium_block, quick_break, transition
Validation plots created	11
CSV output	data/processed/features/ layer2_v02_football_sanity_summary.csv
JSON output	data/processed/features/ layer2_v02_football_sanity_summary.json
Figure output folder	outputs/figures/layer2_sanity/

16.6 Reasonableness Check Results

Reasonableness check	Status	Interpretation
team_width_pitch_bound	warning	At least one team_width value exceeds the current pitch-width reasonableness bound. This should be reviewed before team_width is used for stronger tactical claims. It may reflect true edge/outlier behavior, tolerance settings, broadcast/extrapolated tracking behavior, or a width/depth definition issue that needs football review.
team_depth_pitch_bound	pass	No unreasonable team_depth values were detected by the current bound.

convex_hull_pitch_area_bound	pass	Convex hull area did not exceed the pitch-area reasonableness bound.
negative_metric_values	pass	No impossible negative width/depth/hull-type values were detected.
player_count_used	pass	Player counts used in metric calculations were reasonable for the current SkillCorner tracking outputs.
detection_quality	pass	Detected and extrapolated player counts were documented without triggering a QA failure.
phase_context_presence	pass	Phase context was present for phase-specific validation rows.

Final warning recorded: Reasonableness warning: team_width_pitch_bound.

16.7 Football Sanity Validation Coverage

- Metric distribution checks by major SkillCorner phase labels: build_up, create, finish, transition, quick_break, low_block, medium_block, high_block, and defending_transition where present in the data.
- Metric summaries for team_width, team_depth, convex_hull_area, avg_distance_to_centroid, centroid_x, and centroid_y.
- Pitch-shape validation plots for selected match/frame/team examples across possession and defensive phase labels.
- Phase-segment time-series validation plots showing team_width, team_depth, convex_hull_area, and avg_distance_to_centroid over time.
- Reasonableness checks for pitch-bound metric values, convex hull limits, player_count_used, detected/extrapolated counts, and phase-linking context.

16.8 Plot Outputs Created

The workflow created 11 validation plots under outputs/figures/layer2_sanity/. These plots are intentionally analytical validation visuals. They are not designed as polished dashboards or final reporting graphics.

- Pitch-shape sample plots showing pitch outline, player locations, team centroids, convex hulls when valid, ball location when available, match_id, frame, period, phase label, and team_in_possession_id.
- Phase-segment time-series plots showing whether current metrics move smoothly and plausibly during selected phase segments.

16.9 Football Sanity Findings

- The validation workflow ran successfully across all 10 processed SkillCorner matches.
- The existing Layer 2 outputs were available and inspectable at the expected row volumes.
- Most reasonableness checks passed: team_depth, convex_hull pitch-area bounds, negative metric values, player_count_used, detection quality, and phase context presence.
- The only remaining football sanity warning is team_width_pitch_bound.

- The `team_width` warning means `team_width` should be treated cautiously until the outlier examples are inspected visually and the width/depth definitions are confirmed against the project football convention.
- This warning does not invalidate the Layer 2 v0.1 engineering foundation. It does mean football interpretation should not advance to stronger claims until this warning is reviewed.

16.10 Metrics Considered Safer for Next-Stage Descriptive Profiling

Subject to Lead Developer review, the following metrics appear technically and football-sanity safer for early descriptive profiling because they passed the current QA checks:

- `team_depth`, with current pitch-bound reasonableness status pass.
- `convex_hull_area`, with current pitch-area reasonableness status pass.
- `avg_distance_to_centroid`, as a compact shape-spread descriptor that should still be interpreted descriptively rather than as final tactical truth.
- `player_count_used`, `detected_player_count`, and `extrapolated_player_count` as quality-control/context fields.
- phase-specific summaries where phase context is present and where the metric being interpreted has passed football sanity checks.

16.11 Metrics Needing Caution

- `team_width` needs caution because `team_width_pitch_bound` returned warning.
- `centroid_x` should not be used for strong directional tactical claims until attacking-direction normalization is validated and active.
- `centroid_y` can support coordinate sanity and broad lateral-location checks, but should still be interpreted conservatively.
- Any phase comparison should be treated as descriptive validation/profiling only, not as a final tactical identity claim.

16.12 Orientation and Phase Cautions

- Attacking-direction normalization remains inactive. The orientation module is conservative and should not be treated as a completed tactical normalization layer.
- Current analysis coordinates use Layer 1 metric coordinates.
- Because attacking-direction normalization is not active, `centroid_x` should not be used for strong directional tactical claims yet.
- Frame rows without phase context remain excluded from phase-specific interpretation unless future logic explicitly assigns or fills phase context.

16.13 Recommended Next Step

Recommended next technical step: review the `team_width_pitch_bound` warning before adding new metrics or beginning Layer 3 descriptive profiling. This should include inspecting the generated pitch plots and identifying example rows where `team_width` exceeds the pitch-width bound. If the warning is caused by outlier tolerance or broadcast/extrapolated tracking behavior, document that and decide whether to keep, filter, or flag those rows. If the warning reveals a width/depth semantic mismatch, patch the metric definition before downstream use.

After that review, Layer 3 descriptive profiling may be considered only if the Lead Developer accepts the football sanity findings and confirms which current metrics are safe to profile.

16.14 Future Code-Helper Usage Note

Future code helpers should use this addendum together with the Layer 2 v0.1 handoff, the Layer 1 implementation handoff, the validation handoff, and the architecture document. This addendum explains the football sanity validation workflow and final QA result for current Layer 2 metrics. It does not authorize metric expansion or downstream modeling by itself.

16.15 Scope Guard

- This handoff documents the Layer 2 v0.2 football sanity validation checkpoint only.
- Do not add new Layer 2 metrics from this document alone.
- Do not begin compactness interpretation or transition stability modeling from this document alone.
- Do not begin Layer 3 modeling, clustering, tactical identity modeling, AI interpretation, dashboards, reports, recommendations, or advanced tactical claims from this document alone.
- Before expanding metrics or moving into descriptive profiling, obtain Lead Developer review of the team_width warning and the generated validation plots.